SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks.* *(BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I M.Greeshma(192524195)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Greeshma

Short Answer Test

1) A company needs to transfer a 150 MB design file from a client computer to

a remote server over a network with a bandwidth of 50 Mbps. The Transport

Layer divides the file into 1500-byte segments, while the Data Link Layer

encapsulates each segment into frames by adding 40 bytes of header and

trailer. Calculate the total number of segments and frames transmitted, the total

transmission overhead introduced by the Data Link Layer, and estimate the total

transmission time (ignoring propagation delay). Explain how the Transport and

Data Link layers ensure reliable communication.

Sol..

Given:

- File size = 150 MB = 157,286,400 bytes
- Segment size = 1500 bytes
- Overhead = 40 bytes/frame
- Bandwidth = 50 Mbps

1. Number of Segments

Answer: 104,858 segments

2. Number of Frames

One segment = One frame

Answer: 104,858 frames

3. Total Overhead

Answer: 4,194,320 bytes (≈ 4 MB)

4. Transmission Time

Answer: ≈ 25.8 seconds

2) A Transport Layer protocol uses a sliding window size of 6 frames. Each

frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every

window is lost and must be retransmitted, determine the total number of

retransmissions. Explain how flow control improves communication efficiency.

Sol..

Given:

- Sliding window size = 6 frames
- Frame size = 1024 bytes
- Total frames = 48

1. Number of Transmission Windows

Answer: 8 transmission windows

2. Total Retransmissions

One frame is lost in each window.

Answer: 8 retransmissions

3. Flow Control

Flow control prevents the sender from transmitting data faster than the receiver can process it. This reduces congestion, avoids data loss, and improves communication efficiency.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum

Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a

20-byte IP header, calculate the total number of fragments created and the total

header overhead introduced. Explain how the Network Layer ensures successful

delivery of fragmented packets to the destination.

Sol..

Given:

- IP packet size = 12,000 bytes
- MTU = 1500 bytes

- IP header = 20 bytes

1. Data per Fragment

2. Number of Fragments

So,

Answer: 9 fragments

3. Total Header Overhead

Answer: 180 bytes

Network Layer Function

The Network Layer fragments large packets to fit the MTU, assigns fragment information (ID, offset, and flags), routes the fragments, and enables the destination to reassemble them into the original packet.

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each

frame carries 1024 bytes of user data. If the sender transmits 48 frames,

calculate the number of transmission windows required. If one frame in every

window is lost and must be retransmitted, determine the total number of

retransmissions. Explain how flow control improves communication efficiency.

Sol..

Given:

- Sliding window size = 6 frames
- Frame size = 1024 bytes
- Total frames = 48

1. Number of Transmission Windows

Answer: 8 windows

2. Total Retransmissions

One frame is lost in each window.

Answer: 8 retransmissions

3. Flow Control

Flow control regulates the sender's transmission rate so the receiver is not overloaded. It prevents data loss, reduces congestion, and improves reliable communication.

5. A multimedia application transfers a 600 MB video file. The Session Layer

inserts a synchronization checkpoint after every 60 MB of transmitted data.

During transmission, a failure occurs after 390 MB has been transferred.

Determine the number of checkpoints created before failure and calculate the

amount of data that must be retransmitted after recovery. Explain the importance

of synchronization in the Session Layer.

Sol..

Given:

- Video file size = 600 MB
- Checkpoint every 60 MB
- Failure after 390 MB

1. Number of Checkpoints Before Failure

Checkpoints at:

60, 120, 180, 240, 300, 360 MB

Answer: 6 checkpoints

2. Data to be Retransmitted

Last checkpoint = 360 MB

Failure at = 390 MB

Answer: 30 MB

3. Importance of Synchronization

The Session Layer inserts synchronization checkpoints so that after a failure, communication resumes from the last checkpoint instead of restarting from the beginning. This saves time, bandwidth, and improves reliability.

6. Before transmission, the Presentation Layer compresses a 900 MB file by

40%. The compressed data is encrypted, adding 5% overhead to the

compressed size. Calculate the compressed file size, the encrypted file size, and

the total percentage reduction compared to the original file. Explain how the

Presentation Layer improves transmission efficiency and security.

Sol..

Given:

- Original file size = 900 MB
- Compression = 40%
- Encryption overhead = 5%

1. Compressed File Size

Answer: 540 MB

2. Encrypted File Size

Answer: 567 MB

3. Total Percentage Reduction

Answer: 37% reduction

Presentation Layer Function

The Presentation Layer compresses data to reduce transmission time and bandwidth usage, and encrypts data to ensure confidentiality and secure communication.

7.An FTP application transfers 3 GB of data over a network with an effective

throughput of 120 Mbps. The Application Layer divides the data into requests

of 3 MB each. Calculate the total number of requests generated and estimate the

total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Sol..

Given:

- Data size = 3 GB = 3072 MB
- Request size = 3 MB
- Throughput = 120 Mbps

1. Number of Requests

Answer: 1024 requests

2. Transmission Time

Convert data to bits:

Answer: ≈ 210 seconds

3. Application Layer Function

The Application Layer provides services such as FTP, HTTP, and email, allowing users to access network resources. It ensures reliable file transfer through request handling, authentication, and error reporting.

8.A packet experiences the following processing delays at different OSI layers:

Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data

Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms,

and the transmission delay is 12 ms. Calculate the total end-to-end delay

experienced by the packet and explain how each layer contributes to the

communication process.

Sol..

Given:

- Application Layer = 4 ms
- Transport Layer = 3 ms
- Network Layer = 5 ms

- Data Link Layer = 2 ms
- Physical Layer = 1 ms
- Propagation Delay = 18 ms
- Transmission Delay = 12 ms

1. Total Processing Delay
2. Total End-to-End Delay

Answer: 45 ms

Layer Contributions

- Application Layer: Provides network services to users.
- Transport Layer: Ensures reliable data delivery.
- Network Layer: Routes packets to the destination.
- Data Link Layer: Handles framing and error detection.
- Physical Layer: Transmits data as signals over the medium.

9.A network transfers 80 MB of useful data using frames of 1600 bytes, where

each frame contains 80 bytes of protocol overhead added by different OSI

layers. Calculate the total number of frames required, the total overhead

transmitted, and the transmission efficiency as a percentage. Explain how

protocol overhead affects network performance.

Sol..

Given:

- Useful data = 80 MB = 83,886,080 bytes
- Frame size = 1600 bytes
- Overhead per frame = 80 bytes

1. Useful Data per Frame
2. Number of Frames

Answer: 55,189 frames

3. Total Overhead

Answer: 4,415,120 bytes (≈ 4.21 MB)

4. Transmission Efficiency

Answer: 95%

Protocol Overhead

Protocol overhead includes headers and trailers added by OSI layers for addressing, routing, and error detection. While it ensures reliable communication, excessive overhead reduces available bandwidth and overall network efficiency.

10.A hospital transfers a patient's medical record of 240 MB through an IoT-

enabled healthcare network. The Presentation Layer compresses the data by

30%, the Transport Layer divides the compressed data into 1400-byte

segments, and the Data Link Layer adds 60 bytes of overhead to each frame.

Calculate the compressed file size, the total number of segments transmitted, the

total protocol overhead introduced, and the final amount of data transmitted

across the Physical Layer. Explain how multiple OSI layers work together to

ensure secure and reliable communication.

Sol..

Given:

- Original file size = 240 MB
- Compression = 30%
- Segment size = 1400 bytes
- Data Link overhead = 60 bytes/frame

1. Compressed File Size

Answer: 168 MB

2. Number of Segments

Convert to bytes:

Answer: 125,829 segments

3. Total Protocol Overhead

Answer: 7,549,740 bytes (≈ 7.2 MB)

4. Final Data Transmitted

Answer: ≈ 175.2 MB

OSI Layer Cooperation

- Presentation Layer: Compresses data to reduce size and supports encryption for security.
- Transport Layer: Divides data into segments and ensures reliable delivery.
- Data Link Layer: Adds headers/trailers for framing and error detection.
- Physical Layer: Transmits the final data as electrical/optical/wireless signals.